MILITARY SPECIFICATION

SERVO, ANTI-ICE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the anti-ice servo used in the nuclear biological chemical (NBC) protection system of the M1A1 tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4130
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS
FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation Grades JP-4 and JP-5.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base for Preservation and Operation.
MIL-I-8500	- Interchangeability and Replaceability of Component Parts for Aerospace Vehicles.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-F-16884	- Fuel Oil, Diesel, Marine.
MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration Systems Requirements.
MS33540	- Safety Wiring and Cotter Pinning, General Practice for.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

12336776	- Valve.
12336812	- Air Cycle System Assembly.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish servos which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent servos delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.8.1).

3.2.1 Materials. Magnesium shall not be used (see 4.8.1).

3.2.2 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided unless suitably protected against electrolytic corrosion. Dissimilar metals are specified in MIL-STD-889 (see 4.8.1).

3.3 Design and construction.

3.3.1 Interchangeability. Interchangeability shall be in accordance with MIL-I-8500 (see 4.8.1).

3.3.2 Configuration. The servo shall conform to the requirements of Drawing 12336776 (see 4.8.2).

3.3.3 Safety wiring. If safety wiring is incorporated in the design, it shall conform to the requirements of MS33540 (see 4.8.2).

3.3.4 Weight. The weight of the servo shall not exceed 1.5 pounds (lb) (see 4.8.3).

3.3.5 Reliability. The servo shall be designed to achieve an inherent reliability goal of 68,500 mean miles-between-failure (MMBF) when it is operated in accordance with 3.4.1 through 3.4.5.14 (see 4.8.4).

3.3.6 Maintainability. The servo shall not require preventative maintenance. The mean time-to-repair (MTTR) at the depot level of maintenance shall not exceed 8.0 clock hours (see 4.8.5).

3.3.7 Durability. The servo shall be capable of operating in the Air Cycle System Assembly (Drawing 12336812) for 480 hours under the mission duty cycle spectrum as shown in figures 1 and 2 (see 4.8.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.4 Performance. The servo shall meet the requirements of 3.4.1 through 3.4.4 with an air supply pressure of 35 ± 2 pounds per square inch gage (psig).

3.4.1 Actuation. With a condenser outlet pressure of 1.0 psig and a condenser differential pressure in the range of 1.0 to 2.5 pounds per square inch differential (psid), the servo outlet pressure shall be 0 psig. With a condenser differential pressure of not less than 3 psid, the servo outlet pressure shall be 35 ± 2 psig (see 4.8.7.1).

3.4.2 Reset. With a condenser outlet pressure of 1 psig and decreasing differential pressure, the servo outlet pressure shall be 0 psig at a condenser differential pressure of not greater than 1 psid (see 4.8.7.2).

3.4.3 Internal leakage. With a condenser input and output pressure difference of 0 inches mercury (Hg), the output pressure upstream of a 0.020 inch diameter orifice shall not exceed 0.5 psig (see 4.8.7.3).

3.4.4 External leakage. With a condenser input and output pressure difference of 0 inches Hg, the servo external leakage rate shall not exceed 0.20 pound per minute (see 4.8.7.4).

3.4.5 Environmental. Requirements of 3.4.1 through 3.4.4 shall be met under the following environmental conditions.

3.4.5.1 High temperature. The servo shall meet requirements during and after exposure to an operating ambient air temperature of not less than 125 degrees Fahrenheit (°F), and an inlet air temperature of 290 (+15, -0)°F; except that the external leakage of 3.4.4 shall be not greater than 0.17 pound per minute. The servo shall also meet requirements after exposure to an ambient air storage temperature of not less than 160°F (see 4.8.7.5.1).

3.4.5.2 Low temperature. The servo shall meet requirements during and after exposure to an ambient air temperature of minus 60°F. The servo shall also meet requirements after exposure to a nonoperating storage temperature of minus 60°F (see 4.8.7.5.2).

3.4.5.3 Humidity. The servo shall meet requirements during and after exposure to a relative humidity of up to 100 percent (see 4.8.7.5.3).

3.4.5.4 Sand. The servo shall meet requirements after exposure to sand particles of 0.01 to 1.00 millimeter (mm) diameter and 0.3 ± 0.2 gram per cubic foot density blown against external component surfaces by air with a velocity of 3040 feet per minute (30 knots) (see 4.8.7.5.4).

3.4.5.5 Dust. The servo shall meet requirements after exposure to dust laden air with dust particles of 0.0001 to 0.01 mm diameter and 6 nanograms per cubic centimeter density blown against external component surfaces at an air velocity of 3547 feet per minute (35 knots) (see 4.8.7.5.5).

3.4.5.6 Salt fog. The servo shall meet requirements during and after exposure to a salt fog (see 4.8.7.5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.4.5.7 Chemicals.

3.4.5.7.1 Fuel and hydraulic fluid. The servo shall meet requirements after exposure to the vapors of and while in direct contact with the following materials for 48 hours (see 4.8.7.5.7.1):

Fuel per VV-F-800, MIL-T-5624, MIL-G-3056, and MIL-F-16884;

Hydraulic fluid per MIL-H-6083 and MIL-H-46170.

3.4.5.7.2 Cleaning agents. The servo shall meet requirements during and after exposure to the vapors of and while in direct contact with the following material for 48 hours (see 4.8.7.5.7.2):

Cleaning agents per P-C-437.

3.4.5.8 Cleaning spray. The servo shall meet requirements after exposure to a jet spray of cleaning solution conforming to P-C-437 immediately followed by a cold water rinse (see 4.8.7.5.8).

3.4.5.9 Fungus. The servo shall not support fungus growth when exposed to inoculation by spraying external surfaces with a spore suspension as defined by the "Specimen Inoculation" paragraph of MIL-F-13927, followed by exposure to ambient air temperatures of 80 to 84°F at relative humidities between 96 and 100 percent for a 28-day duration (see 4.8.7.5.9).

3.4.5.10 Vibration. The servo shall meet requirements after exposure to sinusoidal vibration in accordance with table II in each of three mutually perpendicular axes, as illustrated in figure 3 (see 4.8.7.5.10).

TABLE II. Vibration levels.

Axis	Frequency (Hz)	Amplitude
Vertical	5 to 25 25 to 57 57 to 500	± 1 gravity unit 0.030 inch double amplitude (DA) ± 5 gravity units
Longitudinal and lateral	5 to 25 25 to 44 44 to 50	± 1 gravity unit 0.030 inch DA ± 3 gravity units

3.4.5.11 Basic shock. The servo shall meet requirements after exposure to shock impulses of 30 ± 3 gravity units, 11.1 ± 1.1 milliseconds (ms) half-sine wave, applied in each direction of three mutually perpendicular axes, as illustrated in figure 3 (see 4.8.7.5.11).

3.4.5.12 Gun firing shock. The servo shall meet requirements after exposure to shock impulses of 100 ± 10 gravity units, 1.5 ± 0.2 ms half-sine wave, applied in each direction of three mutually perpendicular axes, as illustrated in figure 3 (see 4.8.7.5.12).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.4.5.13 Ballistic shock. The servo shall meet requirements during and after subjection to ballistic shock impulses of 200 ± 20 gravity units, 0.5 ± 0.1 ms half-sine wave, applied in each direction of three mutually perpendicular axes, as illustrated in figure 3 (see 4.8.7.5.13).

3.4.5.14 Submergence. The servo shall meet requirements following submergence in water to a depth of 3 feet for 30 minutes with the inlet and outlet ports capped (see 4.8.7.5.14).

3.4.6 Proof pressure. The servo shall meet 3.4 through 3.4.4 after it is subjected to the following proof pressures.

3.4.6.1 Proof pressure, high temperature. The servo shall be capable of withstanding, without damage or permanent deformation, a proof pressure of 56 ± 5 psig at an inlet air temperature of 290 ± 5°F (see 4.8.7.6.1).

3.4.6.2 Proof pressure, ambient. The servo shall be capable of withstanding, without damage or permanent deformation, a proof pressure of 58 (+2, -0) psig at ambient inlet air temperature and an input air supply pressure of 35 ± 2 psig applied to the shutoff solenoid (see 4.8.7.6.2).

3.4.7 Burst pressure. The servo shall be capable of withstanding, without rupture, a burst pressure of 93 (+10, -0) psig applied at an inlet air temperature of 290 (+15, -0)°F (see 4.8.7.7).

3.5 Identification and marking. All marking and identification plates shall be in accordance with MIL-STD-130 (see 4.8.2).

3.6 Workmanship. Workmanship shall be in accordance with high-grade practices and quality to ensure safety, proper operation, and service (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Test (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F
b. Barometric pressure 28.5 + 2.0 inches Hg
   - 3.0
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply servos that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two servos. Preproduction inspection shall consist of inspection as specified in table III, in the sequence of table IV.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two units from the first ten servos produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table III, in the sequence of table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.4.3 First article inspection failure. Test item deficiencies during or as a result of the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of the first article test shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction of the Government.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Test	Control
Materials and construction	3.2 thru 3.3.1	4.8.1	X			
Defects (see 4.5.2 and table V)	3.3.2, 3.3.3, 3.5 and 3.6	4.8.2	X	X		
Weight	3.3.4	4.8.3	X			
Reliability	3.3.5	4.8.4	X 1/			
Mean repair time	3.3.6	4.8.5	X 1/			
Durability:	3.3.7	4.8.6				
First article		4.8.6.1	X			
Control		4.8.6.2				X
Performance	3.4	4.8.7				
Actuation	3.4.1	4.8.7.1	X		X	
Reset	3.4.2	4.8.7.2	X		X	
Internal leakage	3.4.3	4.8.7.3	X		X	
External leakage	3.4.4	4.8.7.4	X		X	
Environmental	3.4.5	4.8.7.5				
High temperature	3.4.5.1	4.8.7.5.1	X			X
Low temperature	3.4.5.2	4.8.7.5.2	X			X
Humidity	3.4.5.3	4.8.7.5.3	X			
Sand	3.4.5.4	4.8.7.5.4	X			
Dust	3.4.5.5	4.8.7.5.5	X			
Salt fog	3.4.5.6	4.8.7.5.6	X			
Chemicals:	3.4.5.7	4.8.7.5.7				
Fuel and hydraulic fluid	3.4.5.7.1	4.8.7.5.7.1	X			
Cleaning agents	3.4.5.7.2	4.8.7.5.7.2	X			
Cleaning spray	3.4.5.8	4.8.7.5.8	X			
Fungus	3.4.5.9	4.8.7.5.9	X			
Vibration:	3.4.5.10	4.8.7.5.10				
First article		4.8.7.5.10.1	X			
Control		4.8.7.5.10.2				X
Basic shock:	3.4.5.11	4.8.7.5.11				
First article		4.8.7.5.11.1	X			
Control		4.8.7.5.11.2				X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Test	
Gun firing shock:	3.4.5.12	4.8.7.5.12				
First article		4.8.7.5.12.1	X			
Control		4.8.7.5.12.2				X
Ballistic shock	3.4.5.13	4.8.7.5.13	X			
Submergence	3.4.5.14	4.8.7.5.14	X			X
Proof pressure	3.4.6	4.8.7.6				
Proof pressure, high temperature	3.4.6.1	4.8.7.6.1	X			X
Proof pressure, ambient	3.4.6.2	4.8.7.6.2			X	
Burst pressure	3.4.7	4.8.7.7	X			

1/ Only required during preproduction inspection.

TABLE IV. Sequence of tests.

Test sequence	Paragraph	First article sample 1	First article sample 2
Materials and construction	4.8.1	X	X
Defects (see table V)	4.8.2	X	X
Weight	4.8.3	X	X
Proof pressure	4.8.7.6	X	X
Actuation	4.8.7.1	X	X
Reset	4.8.7.2	X	X
Internal leakage	4.8.7.3	X	X
External leakage	4.8.7.4	X	X
High temperature	4.8.7.5.1	X	X
Low temperature	4.8.7.5.2	X	X
Humidity	4.8.7.5.3	X	X
Sand	4.8.7.5.4	X	X
Dust	4.8.7.5.5	X	X
Salt fog	4.8.7.5.6	X	X
Chemicals	4.8.7.5.7	X	X
Cleaning spray	4.8.7.5.8	X	X
Fungus	4.8.7.5.9	X	
Vibration	4.8.7.5.10	X	X
Basic shock	4.8.7.5.11	X	X
Gun firing shock	4.8.7.5.12	X	X
Ballistic shock	4.8.7.5.13	X	X
Submergence	4.8.7.5.14	X	X
Durability	4.8.6	X	X
Reliability 1/	4.8.4	X	X
Mean repair time 1/	4.8.5	X	X
Burst pressure	4.8.7.7	X	X

1/ Only required during preproduction inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the servos of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 20 consecutively produced servos that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	4.0

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table V.

TABLE V. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Incorrect dimensions affecting interchangeability (see 3.3.2).	SIE 1/
102	Improper identification marking (see 3.5).	Visual
Minor	AQL 4.0% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.3.2).	SIE 1/
202	Improper routing of safety wire (see 3.3.3).	Visual
203	Faulty workmanship (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test. Each servo shall be subjected to the quality conformance tests specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.6 Control tests. Control tests shall be conducted on one servo from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a 6-month period, nor less than one test in a 12-month period. The servos shall be subjected to the control tests specified in table III.

4.7 Failure. Failure of any servo to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.1 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3.2, 3.3.3, 3.5, and 3.6 shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Weight. To determine conformance to 3.3.4, weigh the servo.

4.8.4 Reliability. To determine conformance to 3.3.5, the MMBF requirement shall be verified by reliability analysis of test data collected from vehicle or laboratory tests, or both, as specified by the Government. Use of the test and failure data on similar components is permitted providing suitable allowances are made for differences in design, usage, and environment. In the event that test history data are not sufficient to provide satisfactory evidence of the servo MMBF, compliance with the reliability goal shall be determined on the basis of formal rate data sources and prediction procedures.

4.8.5 Maintainability. Conformance to 3.3.6 shall be determined by formal maintainability analysis and predictions, employing recognized and approved failure rate and MTTR data sources and prediction procedures.

4.8.6 Durability.

4.8.6.1 First article. To determine conformance to 3.3.7, subject the servo to 480 hours of operation with conditions simulating the intended usage, including start up, operation, and "off" time.

4.8.6.2 Control. Perform the test of 4.8.6.1, except that the total test time shall be 96 hours.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

4.8.7 Performance. Unless otherwise specified, the servo shall be capable of performing at the pressure level as specified in 3.4. Pressure shall be applied for all tests unless otherwise indicated herein. Throughout the performance tests, install test port fitting A, uncap the outlet port, and install a gauge capable of 0 to 30 pounds per square inch (psi) with a 0.016- to 0.017-inch diameter orifice venting to ambient.

4.8.7.1 Actuation. To determine conformance to 3.4.1, install the servo in the setup of figure 4. Establish the input pressure conditions of 3.4.1 and determine the servo output pressure.

4.8.7.2 Reset. To determine conformance to 3.4.2, install the servo in the setup of figure 4. Establish the input pressure conditions of 3.4.2 and determine the servo output pressure.

4.8.7.3 Internal leakage. To determine conformance to 3.4.3, install the servo in the test setup of figure 4. Establish the inlet conditions of 3.4.3 and measure the servo internal leakage upstream of a 0.020-inch diameter orifice.

4.8.7.4 External leakage. To determine conformance to 3.4.4, install the servo in the test setup of figure 4. Close the downstream shutoff valve to seal off the servo outlet port. Establish the inlet conditions of 3.4.4 and measure the servo external leakage at the inlet port.

4.8.7.5 Environmental. To determine conformance to 3.4.5:

a. Servos undergoing first article inspections (see table III) shall be subjected to the tests of 4.8.7.1 through 4.8.7.4 before, during (where required in the test method), and immediately after each environmental test specified below.

b. Servos undergoing control tests (see table III) shall have passed the quality conformance tests indicated therein. After all environmental tests have been completed, the servos shall be tested only once for the post-environmental test specified.

c. Compliance with the selected performance tests during exposure to environments as specified herein shall provide evidence of conformance to 3.4.1 and 3.4.4.

4.8.7.5.1 High temperature. To determine conformance to 3.4.5.1, subject the servo to the high temperature test for storage before operation, specified in MIL-STD-810, method 501.2, procedure I. Maintain the temperature at 160°F for a period of 48 hours, while insuring the relative humidity is not in excess of 15 percent. At the conclusion of this time, stabilize the servo at no less than 125°F with an inlet air temperature of 290 (+15, -5)°F applied to the servo; and perform the tests of 4.8.7.1 through 4.8.7.4, except verify that the external leakage of 4.8.7.4 is in conformance with 3.4.5.1. Return the servo to 73 ± 18°F and perform the tests of 4.8.7.1 through 4.8.7.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.8.7.5.2 Low temperature. To determine conformance to 3.4.5.2, subject the servo to the low temperature test specified in MIL-STD-810, method 502.2, procedure I. Maintain the temperature at minus 60°F for a period of 24 hours. At the conclusion of this time, return the servo to 73 ± 18°F and perform the tests of 4.8.7.1 through 4.8.7.4. Performance after the test verifies performance during testing.

4.8.7.5.3 Humidity. To determine conformance to 3.4.5.3, subject the servo to five continuous 48-hour cycles in accordance with figure 5. During humidity exposure, subject the servo to the tests of 4.8.7.1 through 4.8.7.4. Prior to these measurements, accumulated moisture may be removed by turning the test item upside down or shaking. Wiping shall not be permitted. Condition the servo for 24 hours at 73°F and 50 ± 10 percent relative humidity. After the humidity exposure, again subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.4 Sand. To determine conformance to 3.4.5.4, test the servo in accordance with MIL-STD-810, method 510.2, procedure II with test requirements as specified in 3.4.5.4, an internal chamber temperature of 73°F, and a relative humidity of less than 22 percent. Following exposure, subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.5 Dust. To determine conformance to 3.4.5.5, test the servo in accordance with MIL-STD-810, method 510.2, procedure I, except that relative humidity shall be less than 22 percent. Particle size shall be 0.0001 to 0.01 mm diameter and 6 nanograms per cubic centimeter, and air velocity shall be 3547 feet per minute (35 knots). Following exposure, subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.6 Salt fog. To determine conformance to 3.4.5.6, place the servo in a test chamber, and subject it to the salt fog test specified in MIL-STD-810, method 509.2, procedure I. The salt fog solution shall be 5 percent by weight of sodium chloride (NaCl) in 95 percent by weight of distilled water. Maintain the temperature in the exposure area between 90 and 95°F. Fog density shall be approximately 3 quarts of salt solution per 10 cubic feet of chamber volume for 24 hours. After the specified 48-hour exposure, remove the servo from the chamber and conduct the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.7 Chemicals.

4.8.7.5.7.1 Fuel and hydraulic fluid. To determine conformance to 3.4.5.7.1, expose the servo to the vapors of and put it in direct contact with the specified chemicals for a period of not less than 48 hours. After exposure, subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.7.2 Cleaning agents. To determine conformance to 3.4.5.7.2, expose the servo to the vapors of and put it in direct contact with the specified chemicals for a period of no less than 48 hours. During and after exposure, subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.8.7.5.8 Cleaning spray. To determine conformance to 3.4.5.8, subject the servo to a cleaning spray test as specified therein. Apply the jet spray perpendicular to and at a distance no closer than five feet from the component surface at a cleaning rate of 1.0 square foot per minute. The water jet shall be derived from a nozzle having a maximum orifice diameter of 0.25 inch and a maximum nozzle pressure of 110 psig. After exposure to the cleaning spray, visually examine the servo externally for signs of deterioration, and subject it to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.9 Fungus. To determine conformance to 3.4.5.9, test the servo in accordance with MIL-STD-810, method 508.3, except the conditions of 3.4.5.9 shall apply. In lieu of the performance of the MIL-STD-810 test, the contractor shall maintain records providing proof with supporting data that the servo is contructed of materials that will not support fungus growth.

4.8.7.5.10 Vibration. To determine conformance to 3.4.5.10, mount the servo using the installed interface configuration for its intended application or equivalent, and subject the servo to the test of MIL-STD-810, method 514.3. Subject the servo to a sinusoidal vibration test in accordance with the test level requirements of table II, herein, for the specified exposure time. Impose the vibration at a logarithmic sweep rate of 15 minutes per sweep while cycling between 5 to 500 to 5 hertz (Hz). During exposure, monitor the servo for intermittent operation. At the conclusion of this test, examine the servo for damage and subject it to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.10.1 First article. The exposure time shall be 60 minutes in each of three orthogonal axes, for a total of 180 minutes.

4.8.7.5.10.2 Control. The exposure time shall be 30 minutes in each of three orthogonal axes, for a total of 90 minutes.

4.8.7.5.11 Basic shock. To determine conformance to 3.4.5.11, mount the servo, using the installed interface configuration for its intended application or equivalent; and subject the servo to a basic design shock pulse test. Apply a total of three half-sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. Peak amplitude shall be as specified. At the conclusion of this test, examine the servo for damage and subject it to the tests of 4.8.7.1 through 4.8.7.4. Performance after the test verifies performance during the test.

4.8.7.5.11.1 First article. Shock pulses shall be 30 $\pm$ 3 gravity units, 11.1 $\pm$ 1.1 ms half-sine wave.

4.8.7.5.11.2 Control. Shock pulses shall be 25 $\pm$ 2.5 gravity units, 11 $\pm$ 1.1 ms half-sine wave.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.8.7.5.12 Gun firing shock. To determine conformance to 3.4.5.12, mount the servo, using the installed interface configuration for its intended application or equivalent; and subject the servo to a basic design shock pulse test. Apply a total of three half-sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. Peak amplitude shall be as specified. At the conclusion of this test, examine the servo for damage and subject it to the tests of 4.8.7.1 through 4.8.7.4. Performance after the test verifies performance during the test.

4.8.7.5.12.1 First article. Shock pulses shall be 100 ± 10 gravity units, 1.5 ± 0.2 ms half-sine wave.

4.8.7.5.12.2 Control. Shock pulses shall be 75 ± 10 gravity units.

4.8.7.5.13 Ballistic shock. To determine conformance to 3.4.5.13, mount the servo, using the installed interface configuration for its intended application or equivalent; and subject the servo to a basic design shock pulse test. Apply a total of three half-sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. Peak amplitude shall be in accordance with 3.4.5.13. At the conclusion of this test, examine the servo for damage and subject it to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.5.14 Submergence. To determine conformance to 3.4.5.14, cover the inlet and outlet ports with waterproof caps. Submerge the servo to a depth of 3 feet for 30 minutes. Measure submergence depth between the water surface and the top surface of the component. After submergence, inspect the servo for damage and subject it to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.6 Proof pressure. At the conclusion of the tests of 4.8.7.6.1 and 4.8.7.6.2, subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.6.1 Proof pressure, high temperature. To determine conformance to 3.4.6.1, install the servo in the test setup of figure 4. Establish the inlet conditions of 3.4.6.1 for no less than 5 minutes. At the completion of the test, verify that the servo has no permanent deformation or damage and subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.6.2 Proof pressure, ambient. To determine conformance to 3.4.6.2, install the servo in the test setup of figure 4. Apply the input power of 3.4 to the shutoff solenoid and establish the inlet conditions of 3.4.6.2 for no less than 5 minutes. At the completion of the test, verify that the servo has no permanent deformation or damage and subject the servo to the tests of 4.8.7.1 through 4.8.7.4.

4.8.7.7 Burst pressure. To determine conformance to 3.4.7, install the servo in the test setup of figure 4. Establish the inlet conditions of 3.4.7 for 5 minutes. Permanent deformation is permissible; however, the servo shall not rupture as a result of the applied pressure.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The anti-ice servo ports supply pressure to the anti-ice valve overriding the torque motor output pressure for condenser anti-icing. The valve is part of the integrated NBC protection system of the M1A1 tank.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Drawing title, number, and date (see 3.3.2).
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.1).
h. If initial production inspection is not required (see 4.4.2).
i. Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Subject term (key word) listing.

Anti-Ice, Servo

6.5 Supersession data. This military specification supersedes General Dynamics specification, SC-12336776, dated 22 March 1984.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

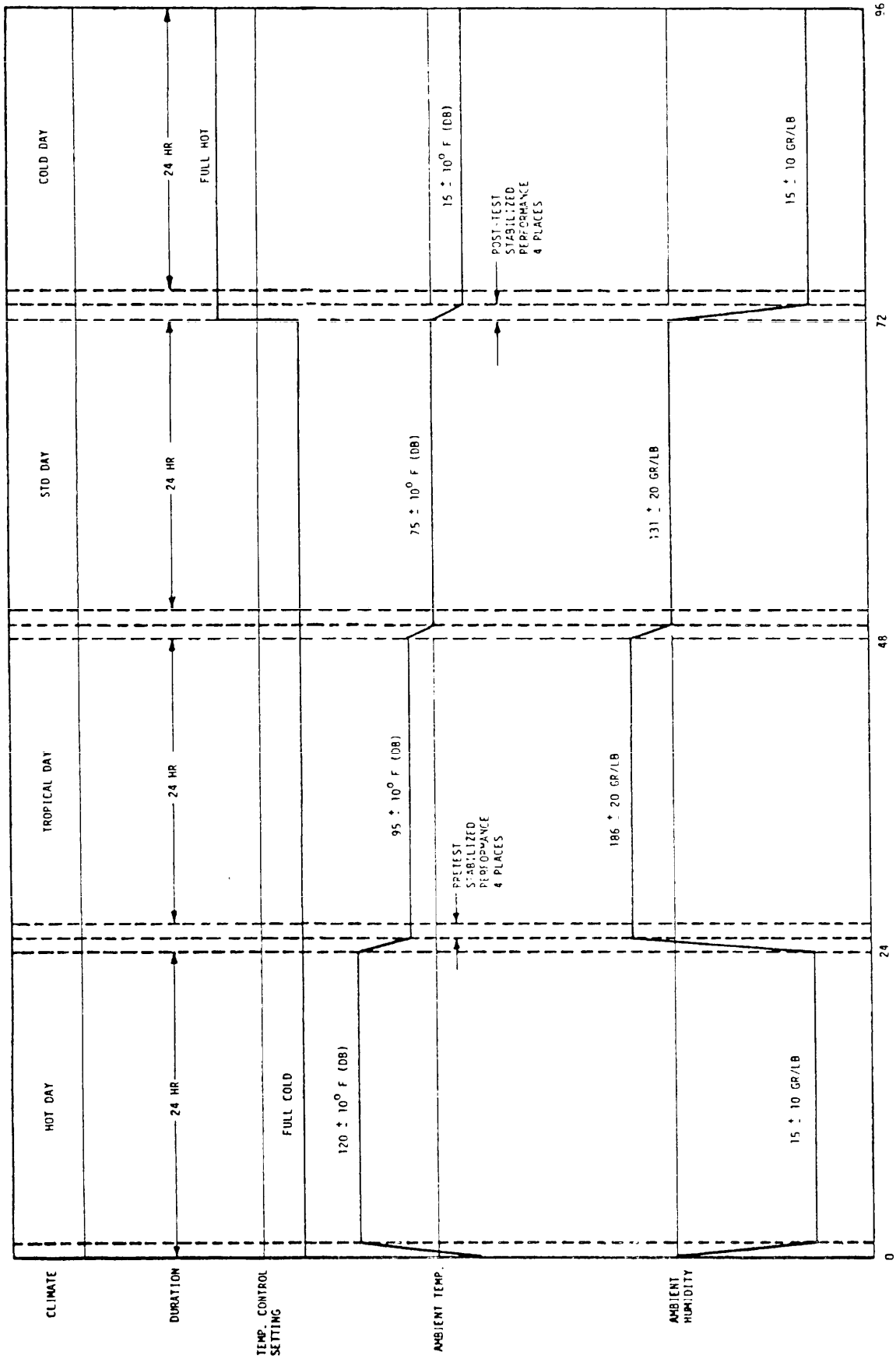


Figure 1. Durability test sequence.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

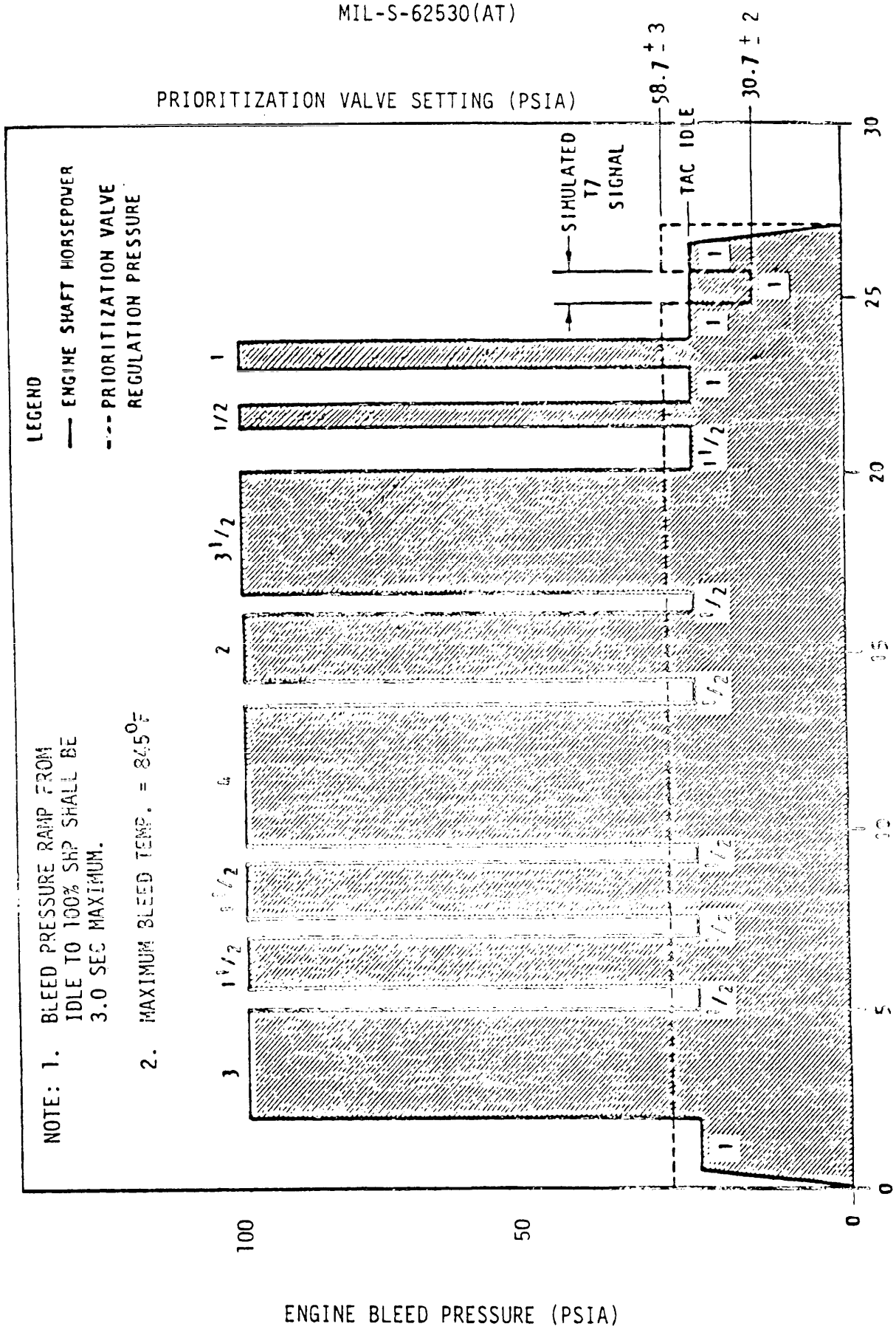


FIGURE 2. Durability test mission duty cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

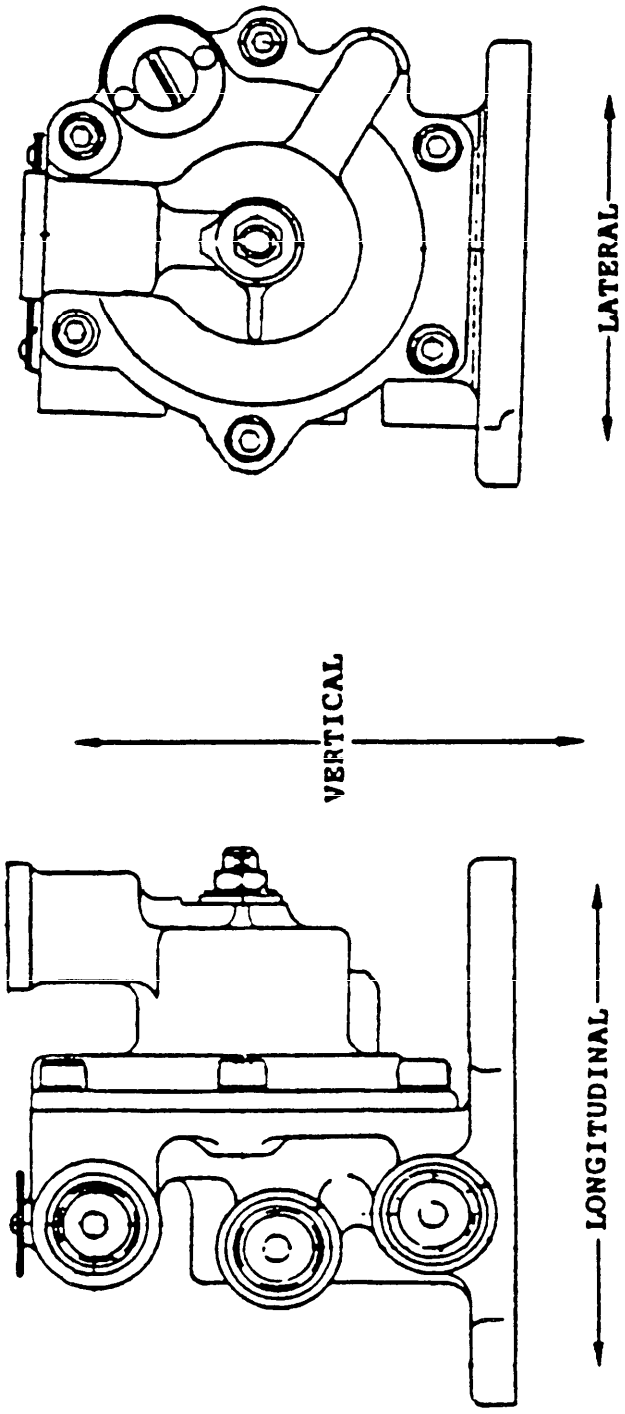


FIGURE 3. Vibration axes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

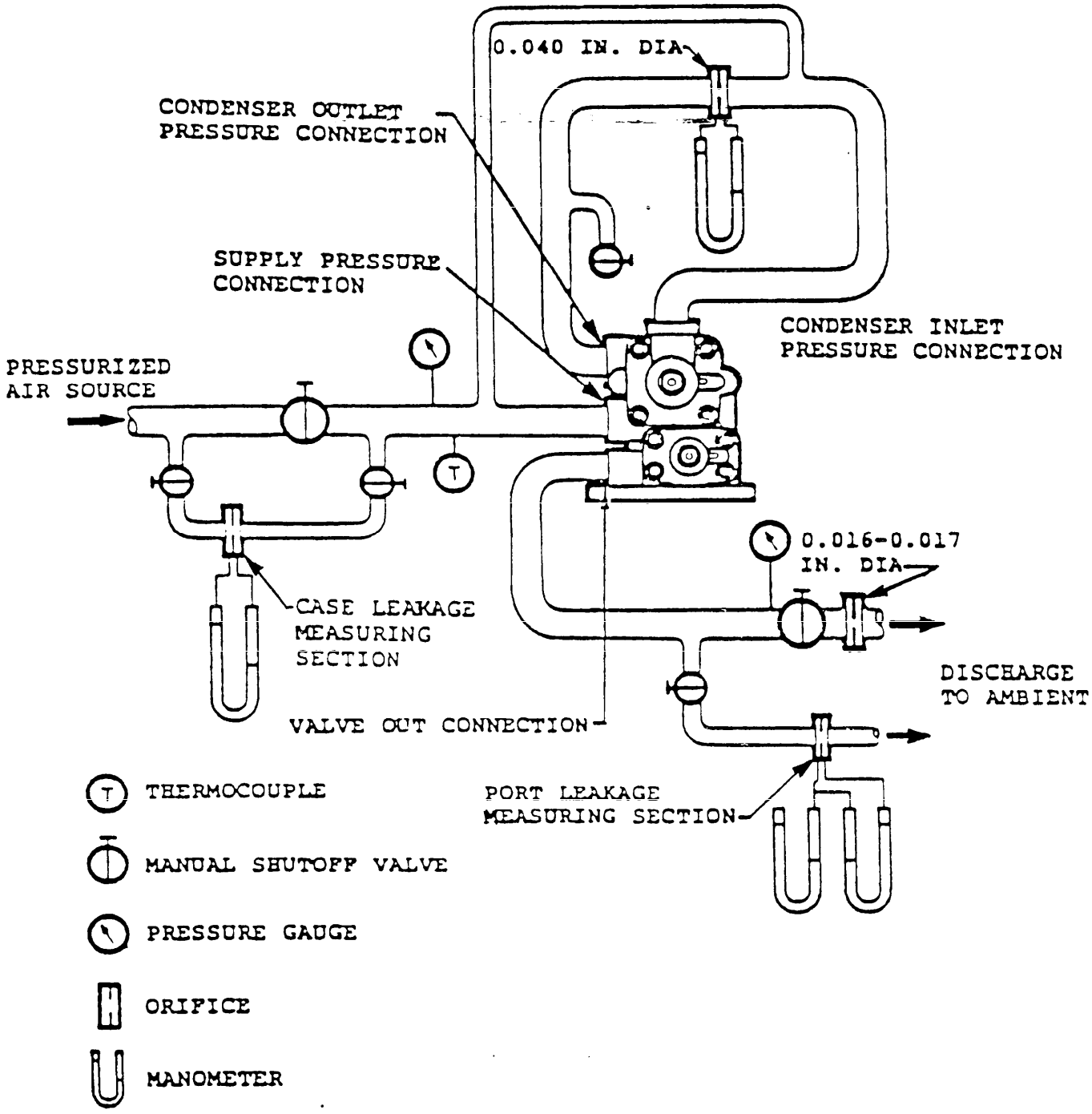
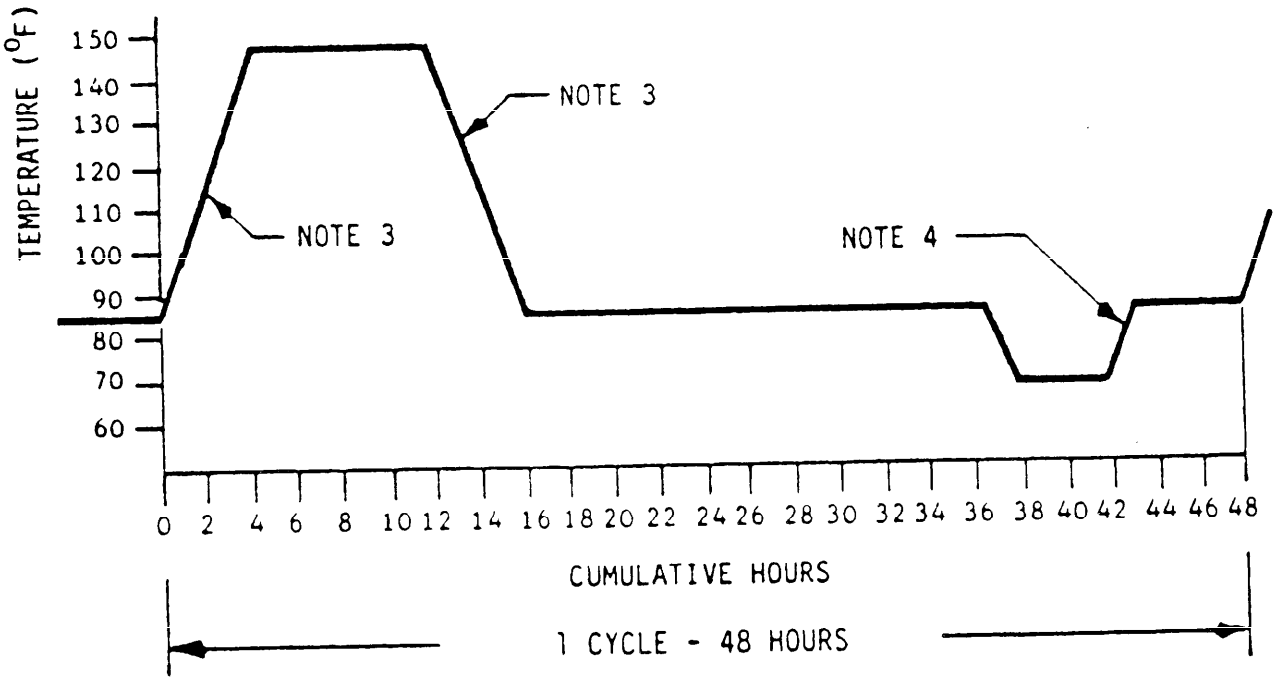


FIGURE 4. Performance and leakage test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.



NOTES:

1. Tolerance during temperature change shall be not greater than $5^{o}F$.
2. Relative humidity shall be maintained at 94 $\pm$4 percent at all times, except that during the descending temperature period the relative humidity may be permitted to drop as low as 85 percent.
3. Rate of temperature change between 86° and $149^{o}F$ shall be not less than $14.4^{o}F$ per hour.
4. The temperature increase in this portion of the curve shall not be less than $18^{o}F$.

FIGURE 4. Humidity cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 4130-A301)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Tank-Automotive Command
ATTN: DRSTA-GSS
Warren, MI 48080

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.